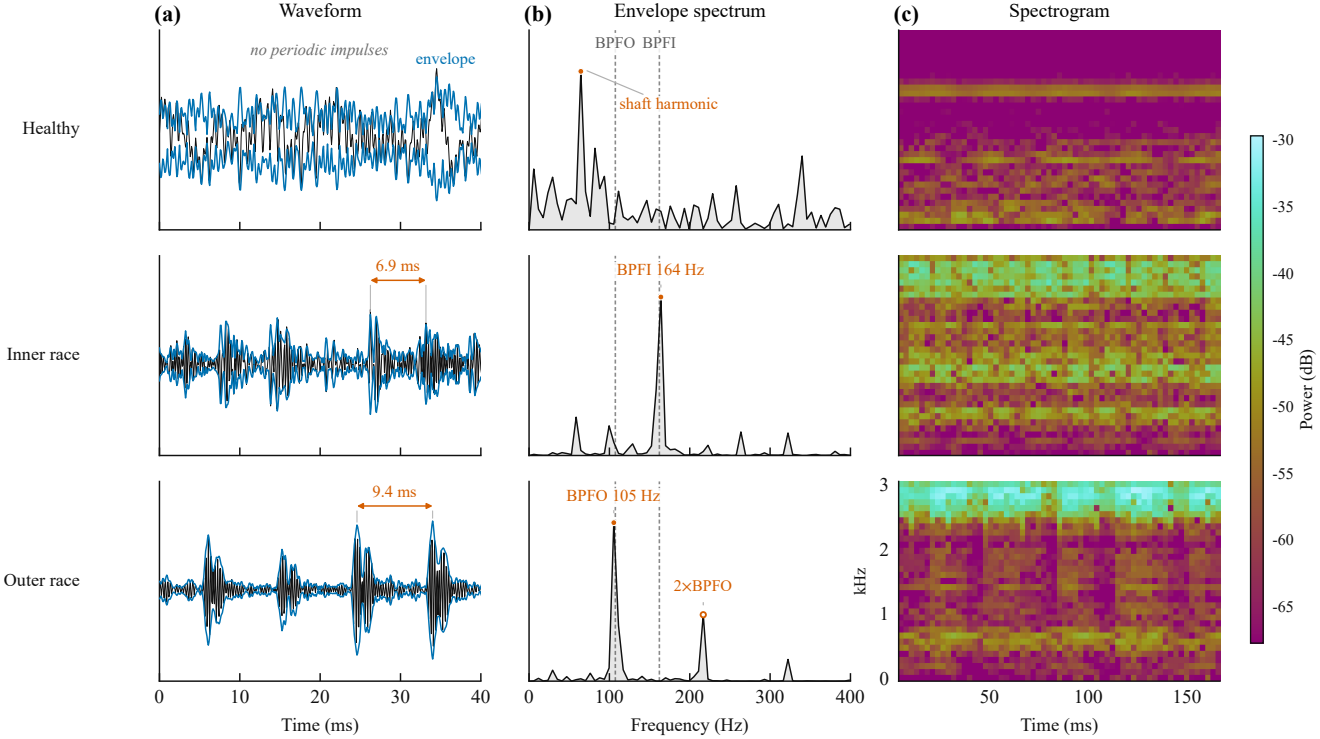

(a)
Waveform
(b)
Envelope spectrum
(c)
Spectrogram
Healthy
Inner race
Outer race
no periodic impulses
envelope
BPFO
BPFI
shaft harmonic
6.9 ms
BPFI 164 Hz
9.4 ms
BPFO 105 Hz
2×BPFO
kHz
Power (dB)
-30
-35
-40
-45
-50
-55
-60
-65
0
1
2
3
0
10
20
30
40
Time (ms)
0
100
200
300
400
Frequency (Hz)
50
100
150
Time (ms)